Tugas 1 Makassar,30 maret 2026

PEMROGRAMAN

BERORIENTASI OBJEK

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PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS ILMU KOMPUTER

UNIVERSITAS MUSLIM INDONESIA

MAKASSAR

2025

Program 1

Tujuan Program
1.Menampilkan nilai dari dua tipe data berbeda (**float** dan **double**) ke layar.
2.Menunjukkan cara deklarasi variabel dan output di Java.

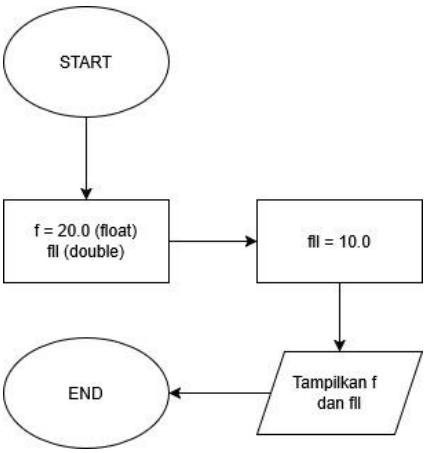
Penjelasan Setiap Bagian
1.class Asgdll → nama kelas.
2.main() → awal program.
3.float f = 20.0f; → variabel pecahan (kecil).
4.double fll; → variabel pecahan (lebih besar).
5.fll = 10.0f; → isi nilai.
6.System.out.println() → tampilkan output.
7.\n → pindah baris.

```
E:\latihan java>javac Asgdll.java

E:\latihan java>java Asgdll.java
f : 20.0
f11: 10.0

E:\latihan java>
```

FLOWCHART



program 2

Tujuan Program
1. Menampilkan teks dan nilai variabel **int**.

Penjelasan Setiap Bagian
1.class Asign → nama kelas.
2.main() → awal program.
3.int i; → deklarasi variabel bilangan bulat.
4.System.out.print("hello\n"); → tampilkan teks + pindah baris.
5.i = 5; → isi nilai variabel.
6.System.out.println() → tampilkan teks + nilai.

```
Microsoft Windows [Version 10.0.26100.7623]
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C:\Users\MSI>E:

E:\>cd latihan java

E:\latihan java>javac Asign.java

E:\latihan java>java Asign.java
hello
Ini nilai i : 5

E:\latihan java>
```
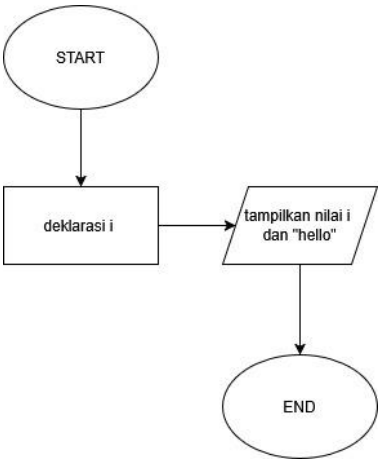



Program 3

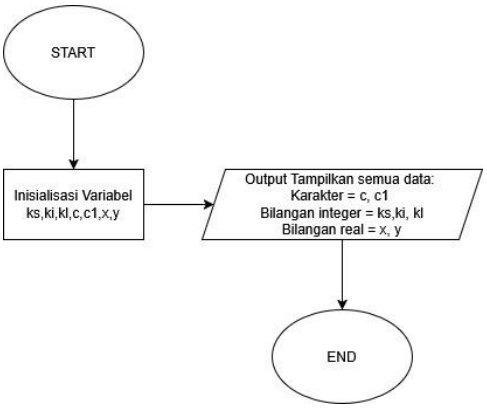
Tujuan Program

Menampilkan berbagai tipe data: **short, int, long, char, float, double**.

Penjelasan Setiap Bagian

1.`short ks` → bilangan kecil.
2. `int ki` → bilangan bulat.
3.`long kl` → bilangan besar.
4.`char c = 65` → karakter dari kode ASCII (**A**).
5.`char c1 = 'Z'` → karakter langsung.
6.`double x` → pecahan lebih presisi.
7.`float y` → pecahan biasa.
8.`System.out.println()` → tampilkan data.
9. `\t` → tab (jarak ke kanan).

```
E:\latihan java> javac ASIGNi.java

E:\latihan java>java ASIGNi.java
Karakter = A
Karakter = Z
Karakter = A
Karakter = Z
Bilangan integer (short) = 1
        (int) = 1
        (long)= 10000
Bilangan Real x = 50.20000076293945
Bilangan Real y = 50.2

E:\latihan java>D
```

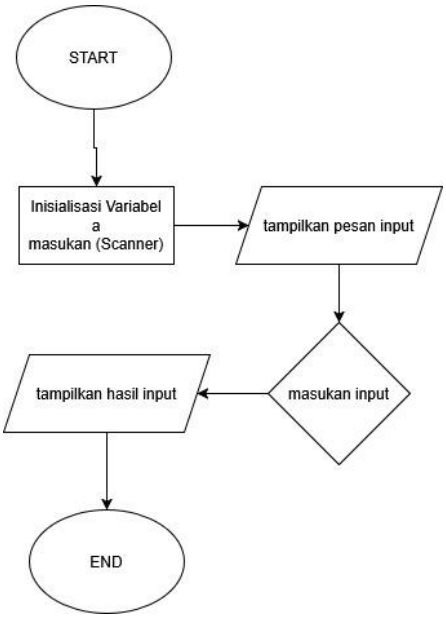


Program 4

Tujuan Program
Membaca input **integer** dari user dan menampilkannya.

Penjelasan Setiap Bagian
1.import java.util.Scanner; → pakai input dari keyboard.
2.Scanner masukan → objek untuk membaca input.
3.new Scanner(System.in) → ambil input dari user.
4.int a; → variabel bilangan bulat.
5.masukan.nextInt() → membaca angka dari user.
6.System.out.print() → tampilkan teks/output.

```
E:\latihan java>javac BacaData.java

E:\latihan java>java BacaData.java
Contoh membaca dan menulis, ketik nilai integer:
10
Nilai yang dibaca : 10
E:\latihan java>
```



Program 5

Tujuan Program

Membaca input **karakter & bilangan** dengan berbagai cara (**BufferedReader, Scanner, Console, JOptionPane**).

Penjelasan Setiap Bagian

1.char cc; int bil; → variabel untuk **karakter & bilangan**
2.InputStreamReader isr → penghubung keyboard
3.BufferedReader dataIn → baca input teks
4.dataIn.readLine().charAt(0) → baca **1 karakter**
5.Integer.parseInt(datAIn.readLine()) → baca **angka**
6.System.out.print() → tampilkan output
7.Scanner sc → input sederhana
8.sc.next().charAt(0) → baca karakter
9.sc.nextInt() → baca angka
10.Console cons → input terminal
11.cons.readLine() → baca input
12.JOptionPane.showInputDialog() → input popup
13.JOptionPane.showMessageDialog() → tampilkan popup

```
Microsoft Windows [Version 10.0.26100.7623]
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C:\Users\MSI>E:

E:\>CD latihan java

E:\latihan java>javac Bacakar.java

E:\latihan java>java Bacakar.java
hello
baca 1 karakter : A
baca 1 bilangan : 10
A
10
bye
Input karakter (Scanner): B
Input bilangan (Scanner): 20
Hasil Scanner: B 20
Input karakter (Console): C
Input bilangan (Console): 30
Hasil Console: C 30

E:\latihan java>_
```
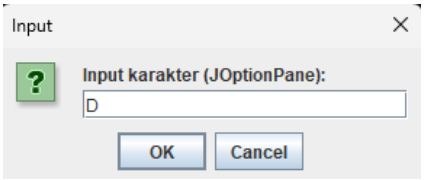
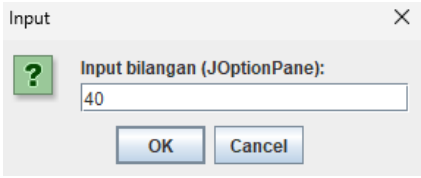
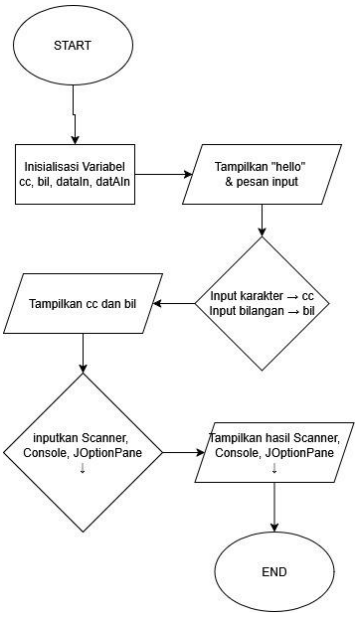





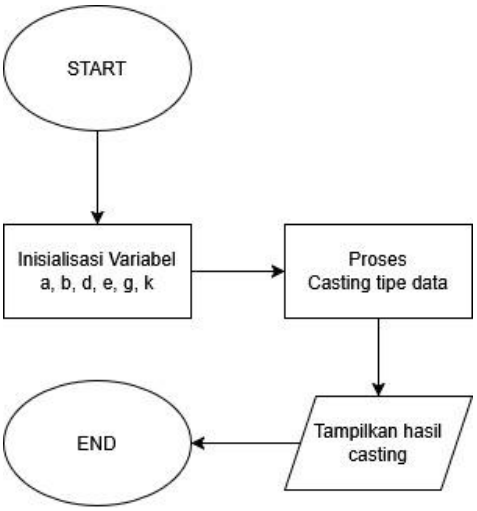


Program 6

Tujuan Program
Menampilkan **casting (konversi tipe data)**

Penjelasan Setiap Bagian
1.int a,b → bilangan bulat
2.float d,e → bilangan pecahan
3.char g → karakter
4.double k → pecahan presisi tinggi
5.(float)a → int → float
6.(double)b → int → double
7.(int)d → float → int (dipotong)
8.(double)e → float → double
9.(int)g → char → angka (ASCII)
10(float)g → char → float
11.(double)g → char → double
12.(int)k → double → int (dipotong)
13.(float)k → double → float
14.System.out.println() → tampilkan hasil

```
E:\latihan java>javac Casting1.java

E:\latihan java>java Casting1.java
5.0
6.0
2
3.200000047683716
53
53.0
53.0
3
3.14

E:\latihan java>_
```



Program 7

Tujuan Program

Menampilkan **casting antar tipe data Class (String & wrapper)**.

Penjelasan Setiap Bagian

1.`String n,m,l` → data dalam bentuk teks

2.`Integer.parseInt(n)` → String → int

3.`Double.parseDouble(m)` → String → double

4.`Float.parseFloat(l)` → String → float

5.`String.valueOf(b)` → int → String

6.`String.valueOf(g)` → char → String

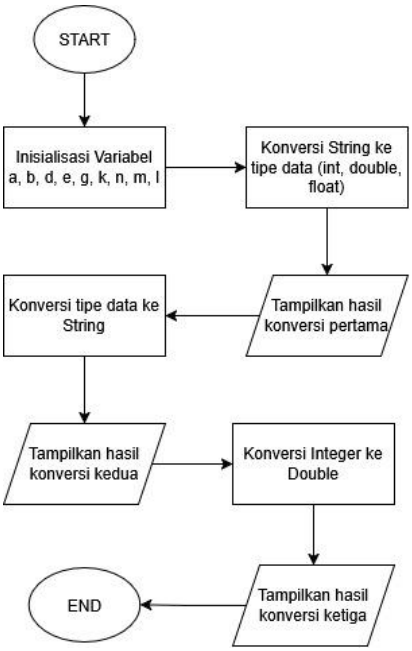
7.`String.valueOf(e)` → float → String

8.`Double.valueOf(a).intValue()` → int → Double → int

9.`Integer.valueOf(b).doubleValue()` → int → double

10.`System.out.println()` → tampilkan hasil

```
E:\latihan java>javac Casting2.java

E:\latihan java>java Casting2.java
a : 67
k : 45.0
d : 100.0
n : 9
m : 5
l : 3.2
k : 67.0
c : 9.0
l : 3.2

E:\latihan java>
```



Program 8

Tujuan Program

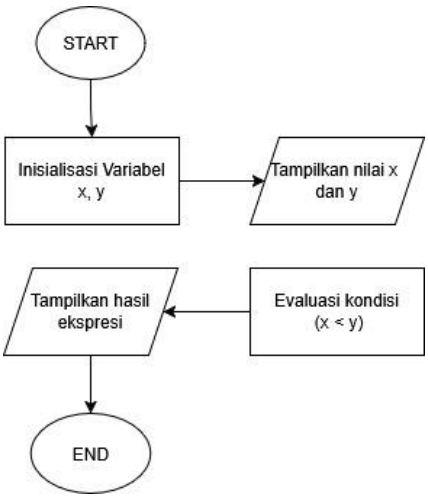
Menentukan nilai dengan **operator kondisional (ternary)**

Penjelasan Setiap Bagian

1.int x=1, y=2 → variabel bilangan
2.System.out.print() → tampilkan nilai
3.(x < y) ? x : y →

- jika **x < y** → ambil x
- jika tidak → ambil y

```
E:\latihan java>javac Ekspresi.java

E:\latihan java>java Ekspresi.java
x = 1
y = 2
hasil ekspresi = (x<y)?x:y = 1
E:\latihan java>
```

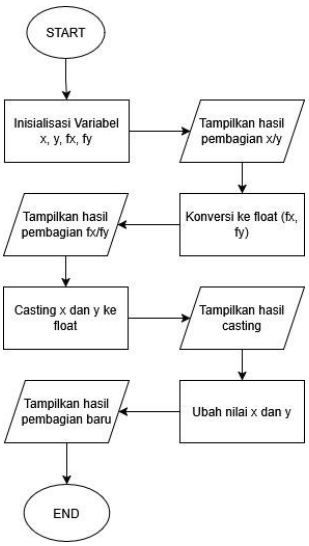


Program 9

Tujuan Program
Menunjukkan **pembagian integer vs float** dan **casting**.

Penjelasan Setiap Bagian
1.int x,y → bilangan bulat
2.float fx,fy → bilangan pecahan
3.x/y → pembagian **integer (hasil dibulatkan)**
4.fx/fy → pembagian **float (ada desimal)**
5.fx = x; fy = y; → ubah ke float
6.(float)x/(float)y → casting langsung ke float

```
E:\latihan java>Javac Ekspresi1.java

E:\latihan java>java Ekspresi1.java
x/y (format integer) = 0
x/y (format float) = 0
x/y (format integer) = 0.5
x/y (format float) = 0.5
float(x)/float(y) (format integer)=0.5
float(x)/float(y) (format float) = 0.5
x/y (format integer) = 3
x/y (format float = 3
E:\latihan java>
```



Program 10

Tujuan Program

Menampilkan teks ke layar dengan **print** dan **println**

Penjelasan Setiap Bagian

1.class PrintHello → nama kelas
2.main() → awal program
3.System.out.print("Hello") → tampil tanpa pindah baris
4.\n → pindah baris
5.System.out.println() → tampil + otomatis pindah baris

```
Microsoft Windows [Version 10.0.26100.7623]
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C:\Users\MSI>E:

E:\>CD latihan java

E:\latihan java>javac PrintHello.java

E:\latihan java>java PrintHello.java
Hello
Hello World
Welcome

E:\latihan java>
```
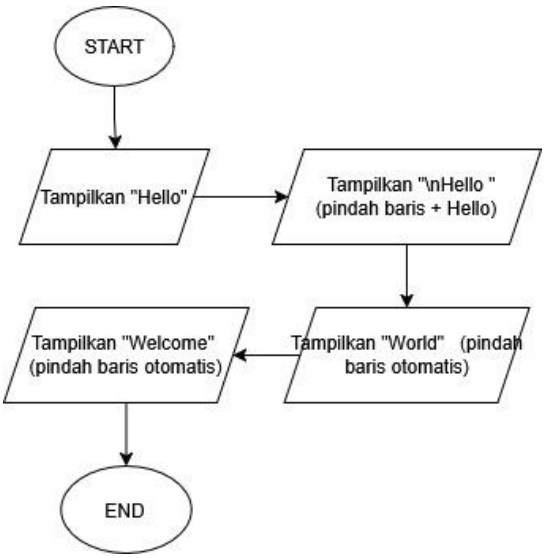



Program 11

Tujuan Program

Menunjukkan efek **operator increment (++)**.

Penjelasan Setiap Bagian

1.int i, j → variabel bilangan
2.i = 3 → nilai awal
3.j = i++ →

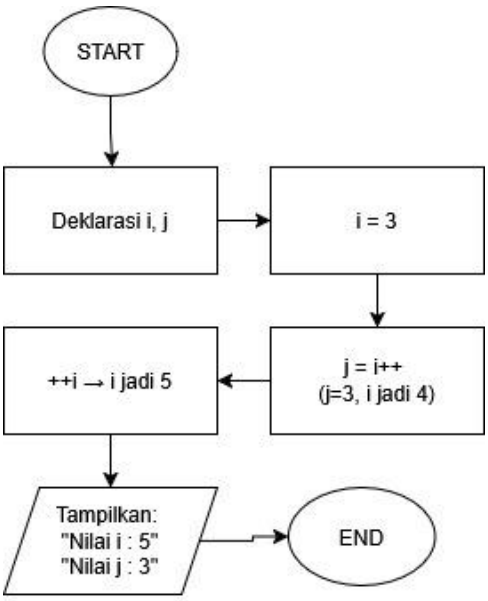
- **post-increment** → nilai i dipakai dulu, baru ditambah
- jadi j = 3, lalu i = 4

4.++i →

- **pre-increment** → ditambah dulu, baru dipakai
- jadi i = 5

5.System.out.println() → tampilkan hasil

```
E:\latihan java>javac  Incr.java

E:\latihan java>java  Incr.java
Nilai i : 5
Nilai j : 3

E:\latihan java>_
```

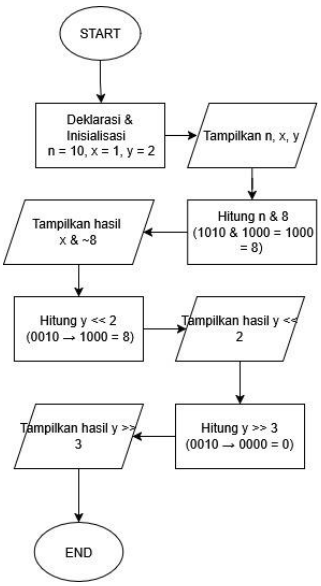


Program 12

Tujuan Program
Menampilkan operasi **bitwise (&, ~, <<, >>)**.

Penjelasan Setiap Bagian
1.int n=10 → biner **1010**
2.int x=1 → biner **0001**
3.int y=2 → biner **0010**
4.n & 8 → AND → **1010 & 1000 = 1000 (8)**
5.x & ~8 → NOT lalu AND → **0001 & 0111 = 0001 (1)**
6.y << 2 → geser kiri → **2 jadi 8**
7.y >> 3 → geser kanan → **jadi 0**
8.System.out.println() → tampilkan hasil

```
E:\latihan java>javac Oper1.java

E:\latihan java>java Oper1.java
n = 10
x = 1
y = 2
n & 8 = 8
x & ~ 8 = 1
y << 2 = 8
y >> 3 = 0

E:\latihan java>
```

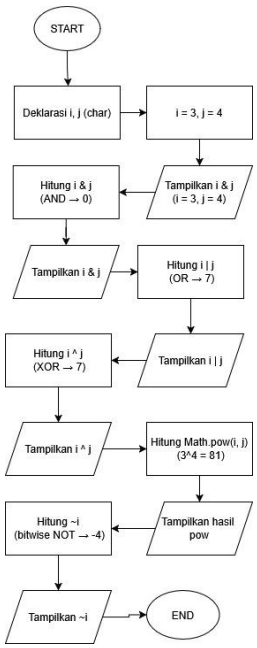


Program 13

Tujuan Program
Menampilkan operasi **bitwise & relasional pada char**.

Penjelasan Setiap Bagian
1.char i=3, j=4 → disimpan sebagai **nilai ASCII (angka)**
2.(int)i → tampilkan nilai angka dari char
3.i & j → AND → hasil **0**
4.i | j → OR → hasil **7**
5.i ^ j → XOR → hasil **7**
6.~i → NOT → hasil **-4**
7.Math.pow(i,j) → pangkat (**3^4 = 81**)
8.^ di Java → **XOR, bukan pangkat**
9.System.out.println() → tampilkan hasil

```
E:\latihan java>javac Oper2.java

E:\latihan java>java Oper2.java
i = 3
j = ♦
i & j = 0
i | j = 7
i ^ j = 7
81.0
~i = -4

E:\latihan java>
```

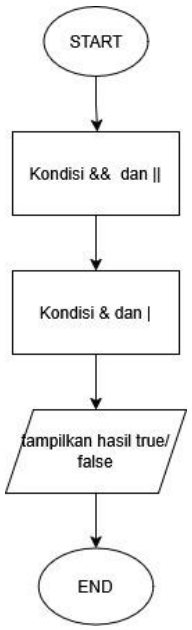


Program 14

Tujuan Program
Menunjukkan perbedaan **operator logika (&&, ||)** dan **bitwise (&, |)** pada boolean.

Penjelasan Setiap Bagian
1.&& → AND (logika, short-circuit)
2.|| → OR (logika)
3.& → AND (bitwise, tetap diproses semua)
4.| → OR (bitwise)
5.if (true && true) → hasil **true**
6.if (true & true) → tetap true, tapi semua dieksekusi
7.if (true || true) → **true**
8.if (true | false) → **true**
9.System.out.println() → tampilkan hasil

```
E:\latihan java>javac Oper3.java

E:\latihan java>java Oper3.java
true
false
true
true
true

E:\latihan java>_
```

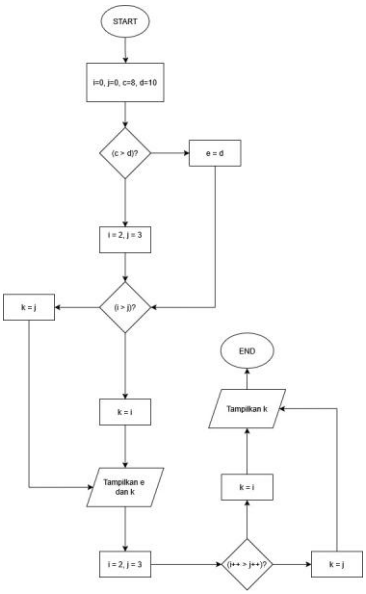


Program 15

Tujuan Program
Menampilkan penggunaan **operator ternary (? :)**.

Penjelasan Setiap Bagian
1.? : → pengganti **if-else**
2.(kondisi) ? nilai1 : nilai2
3.c=8, d=10 → dibandingkan → ambil yang **lebih besar (10)**
4.i=0, j=0 → hasil k = 0
5.i++ , j++ → **post-increment** (dipakai dulu baru ditambah)
6.(i++ > j++) → bandingkan dulu, baru nilai bertambah
7.System.out.print() → tampilkan hasil

```
E:\latihan java>javac Oper4.java

E:\latihan java>java Oper4.java
Nilai e = 10
Nilai k = 0
Nilai k = 4
E:\latihan java>_
```



Program 16

Tujuan Program

Menampilkan hasil **operator boolean, aritmatika, dan relasional**.

Penjelasan Setiap Bagian

1.boolean Bool1, Bool2 → data true/false
2.int i,j → bilangan bulat
3.float x,y → bilangan pecahan

Boolean

- && → AND
- || → OR
- ! → NOT
- ^ → XOR

Aritmatika

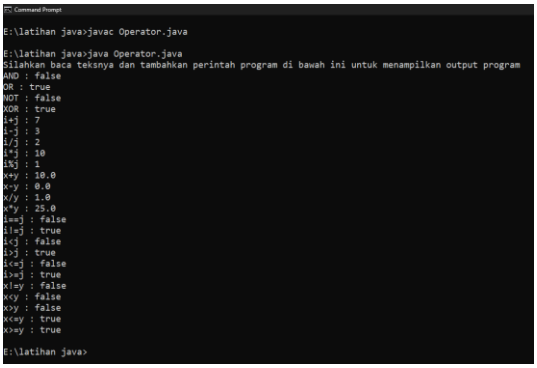
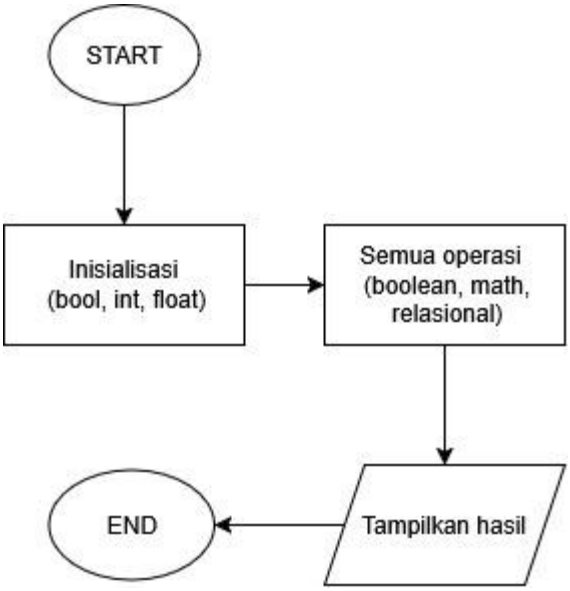
- + - * / % → operasi hitung
- i/j → pembagian **integer (tanpa desimal)**

Float

- x/y → pembagian **desimal**

Relasional

- == != < > <= >= → perbandingan (hasil boolean)



```
Command Prompt

E:\latihan java>javac Operator.java

E:\latihan java>java Operator.java
Silahkan baca teksnya dan tambahkan perintah program di bawah ini untuk menampilkan output program
AND : false
OR : true
NOT : false
XOR : true
i+j : 7
i-j : 3
i/j : 2
i*j : 10
i%j : 1
x+y : 10.0
x-y : 0.0
x/y : 1.0
x*y : 25.0
i==j : false
i!=j : true
i<j : false
i>j : true
i<=j : false
i>=j : true
x!=y : false
x<y : false
x>y : false
x<=y : true
x>=y : true

E:\latihan java>
```